



Barker College

Mathematics

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Student Number

2016
TRIAL
HIGHER SCHOOL
CERTIFICATE

AM Friday 5th August

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☐ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ☒ (B) ☒ ^{*correct*} (C) ☐ (D) ☐

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Start
Here →

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐



Barker College

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Student Number

2016
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Mathematics

Staff Involved:

- RMH* • JGD*
- DZP • MRB
- AXD • LAK
- AJD • KJL
- GPF • ARM

120 copies

General Instructions

- Reading time – 5 minutes
- Working time - 3 hours
- Write using black pen.
- A Reference Sheet is provided
- Approved calculators and Mathaids may be used.
- Diagrams are not to scale unless indicated.
- Marks may not be awarded for careless or badly arranged working.
- In Questions 11 – 16, show all relevant mathematical reasoning and / or calculations

AM Friday 5th August

Total marks – 100

Section I Pages 4 - 7

10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II Pages 9 - 18

90 marks

- Attempt Questions 11 – 16
- Start each question in a NEW booklet.
- Allow about 2 hours 15 minutes for this section

Section I – Multiple Choice (10 marks)

Attempt questions 1 – 10

All about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1 – 10.

1. What is 75680241 written in scientific notation, correct to 3 significant figures?

(A) 7.568×10^7 (B) 7.57×10^7 (C) 7.568×10^4 (D) 7.57×10^5

2. All students in a particular TAFE course sit a theory test and a practical test. 65% of students pass the theory test and only 40% pass the practical test. A student is chosen at random. The probability that the student passes both tests is:

(A) 0.26

(B) 1.05

(C) 0.026

(D) 2.6

3. The solution to $2^{3x+5} = 4^{x-1}$ is:

(A) $x = -2$

(B) $x = -1$

(C) $x = -6$

(D) $x = -7$

4. The parabola $y^2 = 12x - 24$ has:

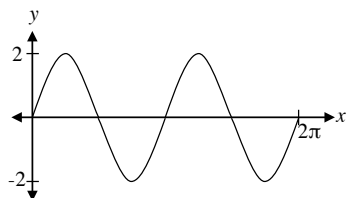
(A) focus (5,0) and directrix $x = -1$

(B) focus (2,3) and directrix $y = -3$

(C) focus (-1,0) and directrix $x = 5$

(D) focus (27,0) and directrix $x = 21$

5.



The graph above could have as its equation:

- (A) $y = 2 \cos \frac{x}{2}$
- (B) $y = 2 \cos x$
- (C) $y = 2 \sin x$
- (D) $y = 2 \sin 2x$

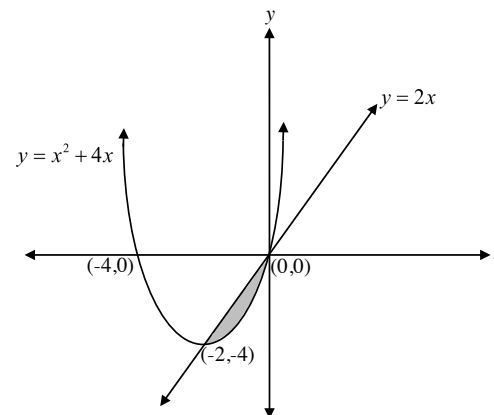
6. What is the value of the derivative of $y = \tan x - 3 \sin 2x$ at $x = 0$?

- (A) 0
- (B) 7
- (C) -5
- (D) -2

7. The domain and range of $y^2 = 4 - x^2$ are:

- (A) $-2 \leq x \leq 2, 0 \leq y \leq 2$
- (B) $-2 \leq x \leq 2, -2 \leq y \leq 2$
- (C) all real $x, y \leq 2$
- (D) $x \leq 4, \text{ all real } y$

8. The diagram shows the parabola $y = x^2 + 4x$ meeting the line $y = 2x$ at $(-2, -4)$ and $(0, 0)$.



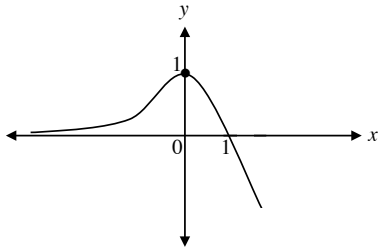
Which expression gives the area of the shaded region bounded by the parabola and the line?

- (A) $\int_{-2}^0 (-2x - x^2) dx$
- (B) $\int_0^{-2} (-2x - x^2) dx$
- (C) $\int_{-2}^0 (x^2 + 2x) dx$
- (D) $\int_0^{-2} (x^2 + 2x) dx$

9. The sum of the first three terms of a geometric series is 19 and the sum to infinity is 27. The values of the first term a , and the common ratio r are:

- (A) $a = -9, r = \frac{-2}{3}$
- (B) $a = 9, r = \frac{2}{3}$
- (C) $a = \frac{19}{3}, r = \frac{19}{27}$
- (D) $a = -9, r = \frac{2}{3}$

10. The diagram shows the graph of $y = e^x(1-x)$.



How many solutions are there to the equation $e^x(1-x) = x^2 - 1$?

- (A) 0
- (B) 1
- (C) 2
- (D) 3

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End of Section I

Section II

90 marks

Attempt Questions 11 - 16

Allow about 2 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

[START A NEW BOOKLET]

(a) Simplify $16a - (3 - 4a)$. 1

(b) Factorise fully $8x^3 - 27$. 2

(c) Express $\frac{3}{2 - \sqrt{5}}$ with a rational denominator. 2

(d) Find $\lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x - 3}$. 2

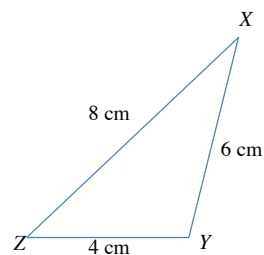
(e) Differentiate $y = \frac{2x}{(e^x + 1)^3}$. 2

(f) The roots of the equation $x^2 - 7x + 9 = 0$ are α and β .
Find the value of $\alpha^2 + \beta^2$. 2

(g) Differentiate $y = x^2 \cos x$. 2

(h) The diagram shows $\triangle XYZ$ with sides
 $XY = 6$ cm, $YZ = 4$ cm and $XZ = 8$ cm. 2

Calculate $\angle XYZ$ to the nearest degree.

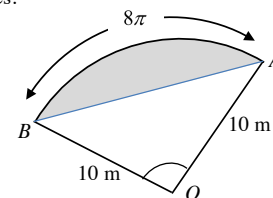


End of Question 11

Question 12 (15 marks)

[START A NEW BOOKLET]

- (a) In the diagram, AB is an arc of a circle with centre O . The radius is 10 metres and the arc length is 8π metres.



(i) Show that $\angle BOA$ is $\frac{4\pi}{5}$ radians. 1

(ii) Show that the area of sector AOB is 40π . 1

(iii) Hence, or otherwise, calculate the area of the shaded segment to the nearest whole number. 1

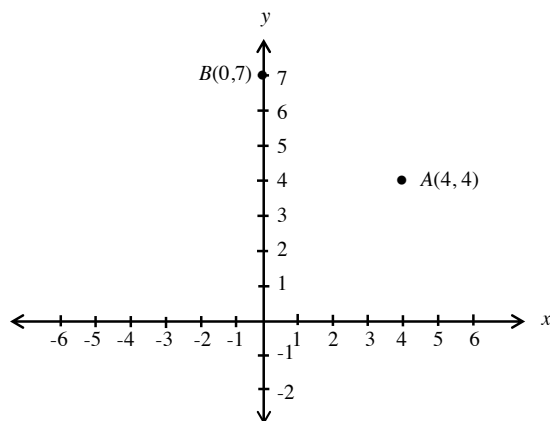
(b) Find the values of k for which the equation $x^2 + kx + 9 = 0$ has two real distinct roots. 2

(c) Solve $2\sin^2 \theta + \sin \theta = 0$ for $0 \leq \theta \leq 2\pi$. 3

Question 12 continues on page 11

Question 12 (continued)

- (d) The diagram below shows the points $A(4, 4)$ and $B(0, 7)$.



- | | |
|---|----------|
| (i) Find the length of AB . | 1 |
| (ii) Find the gradient of AB . | 1 |
| (iii) Show that the equation of AB is $3x + 4y - 28 = 0$. | 1 |
| (iv) Given the point $D(-1, 1)$, find the perpendicular distance from D to the line AB . | 2 |
| (v) Find the coordinates of the point C such that $ABCD$ is a parallelogram. | 1 |
| (vi) Find the area of $ABCD$. | 1 |

End of Question 12

Question 13 (15 marks)

[START A NEW BOOKLET]

- | | |
|--|----------|
| (a) (i) Find the derivative of $y = 3e^{x^2+1}$. | 2 |
| (ii) Hence, or otherwise, find $\int x e^{x^2+1} dx$. | 1 |
| (b) Find the equation of the curve $y = f(x)$ given that the curve has a turning point at $x = 1$, $f'(x) = 3x^2 - 6x + c$ and $f(2) = 7$. | 3 |
| (c) A bag contains 10 blue counters, 8 red counters and 5 green counters. If two counters are drawn from the bag and the first is not replaced, find the probability that: | |
| (i) the second counter drawn is green, given that the first counter drawn is blue. | 1 |
| (ii) both of the counters are blue. | 1 |
| (iii) both counters are the same colour. | 2 |
| (d) Jack borrows \$300 000 to buy a unit. Interest is calculated monthly at the rate of 4.8% p.a compounded monthly. He agrees to repay the loan with equal monthly instalments of \$ M at the end of each month for 15 years. Let A_n be the amount owing after n months. | |
| (i) Find an expression for A_1 . | 1 |
| (ii) Show that $A_{180} = 300\,000(1.004)^{180} - M(1 + 1.004 + \dots + 1.004^{179})$. | 2 |
| (iii) Calculate the amount of the monthly repayment, to the nearest cent. | 2 |

End of Question 13

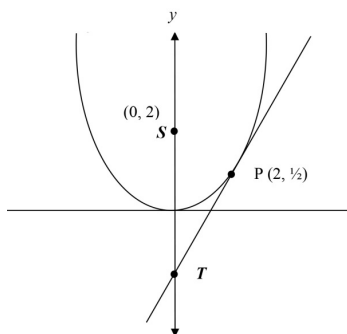
Question 14 (15 marks)

[START A NEW BOOKLET]

(a) Evaluate $\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \sec^2 2x \, dx$. 3

(b) Calculate the area bounded by the curve $y = x^2 - 7x + 10$ and the x -axis. 3

(c) The diagram shows the parabola $x^2 = 8y$ with focus $S(0, 2)$. A tangent to the parabola is drawn at $P(2, \frac{1}{2})$.



(i) Find the gradient of the tangent at point P . 1

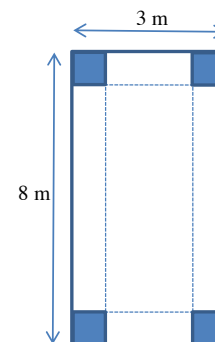
(ii) The tangent at P cuts the y -axis at T . Find the coordinates of T . 1

(iii) Calculate the acute angle (to the nearest minute) that the tangent at P makes with the y -axis. 2

(iv) Calculate the area of $\triangle PTS$. 1

Question 14 (continued)

(d) A rectangular sheet of metal is 8 m by 3 m. Four equal squares, side x m, are removed from each corner. The edges are then turned up to form a box, open at the top.



(i) Show that the volume of the box is given by $V = 4x^3 - 22x^2 + 24x$. 1

(ii) Find the value of x which makes this volume a maximum. 3

End of Question 14

Question 14 continues on page 14

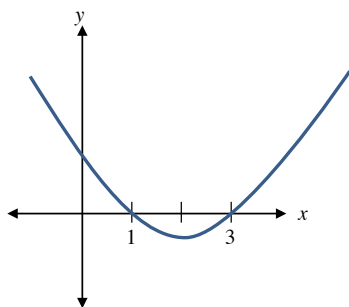
Question 15 (15 marks)

[START A NEW BOOKLET]

(a) (i) Find $\frac{dy}{dx}$ of $y = \log_e(2x-1)$. 1

(ii) Hence, find the value of x when the gradient of the curve is $\frac{2}{5}$. 2

(b) The graph of $y = f'(x)$ is drawn below.



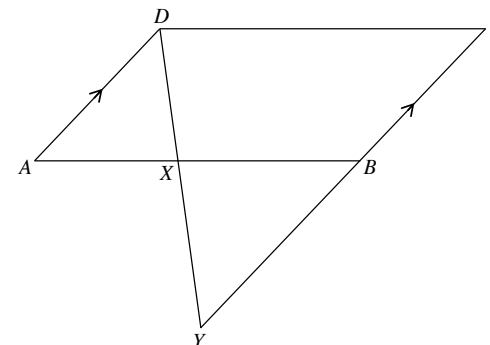
Draw a possible sketch of $y = f(x)$ clearly showing the x -coordinates of any stationary points or points of inflexion. 2

(c) Find $\int \frac{x^2}{x^3+4} dx$ 2

Question 15 (continued)

(d) Use Simpson's Rule, with 5 function values, to estimate the area between the curve $y = \frac{2}{x^2-1}$ and the x -axis from $x = 2$ to $x = 6$ (answer to 3 decimal places). 3

(e) $ABCD$ is a parallelogram.
 X lies on AB .
 DX and CB are both produced to Y .



(i) Prove $\triangle ADX$ is similar to $\triangle CYD$. 3

(ii) Determine the length of XY given that $AX = 8$ cm, $DC = 12$ cm and $DX = 10$ cm. 2

Question 15 continues on page 16

End of Question 15

Question 16 (15 marks) [START A NEW BOOKLET]

(a) Consider the equation $y = (x-2)^3(x+1)$.

(i) Show that $\frac{dy}{dx} = (x-2)^2(4x+1)$.

1

(ii) Given that $\frac{d^2y}{dx^2} = 6(x-2)(2x-1)$ (DO NOT PROVE THIS),

find the coordinates of any stationary points and determine their nature.

3

(iii) Determine the coordinates of any points of inflection.

2

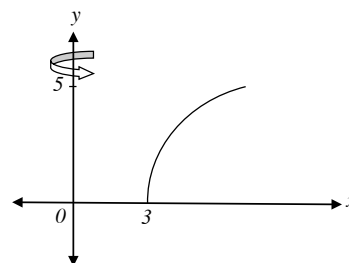
(iv) Hence, sketch the curve $y = (x-2)^3(x+1)$ showing intercepts on the axes and any information from Parts (i) to (iii).

2

Question 16 continues on page 18

Question 16 (continued)

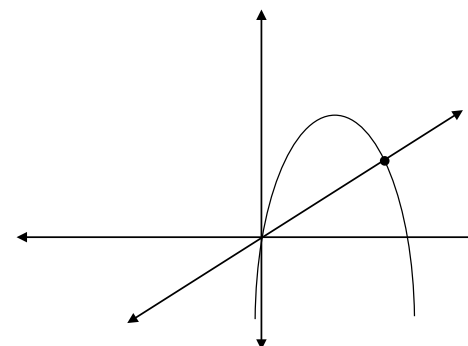
(b) A bowl is formed by rotating the curve $y = 5 \ln(x-2)$ about the y axis for $0 \leq y \leq 5$.



Find the volume of the bowl, giving your answer as a simplified exact value.

3

(c) The graphs of $y = mx$ and $y = 6x - x^2$ intersect at the origin and at point B .



Find the area, in simplest form in terms of m , bounded by $y = mx$ and $y = 6x - x^2$.

4

End of Question 16

End of Paper

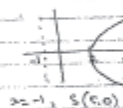
Section I. Multiple choice

1. 7.57×10^7

2. $0.65 \times 10^4 = 6500$

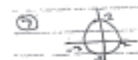
3. $\int_{-2}^2 x^2 dx$
 $= \frac{x^3}{3} \Big|_{-2}^2$
 $= \frac{8}{3} - \left(-\frac{8}{3}\right) = \frac{16}{3}$

4. $y^2 = 12(x-2)$
 $\therefore 12 = 12$
 $a = 3, \sqrt{12}$



5. $y = 2 \sin 2x$

6. $y' = \sin^2 x - 3 \sin 2x$
 $\text{At } x=0, y' = 1 - 0 = 1$



7. $\int_0^{\pi} 2x \cos(x^2 + \pi) dx$
 $= \int_0^{\pi} (2x \cos x^2 - 2x \sin x^2) dx$

1. $\frac{a(1-r^n)}{1-r} = 19$

2. $\frac{a}{1-r} = 27$
 $\therefore 27(1-r) = 19$
 $27 - 27r = 19$
 $-27r = -8$
 $r = \frac{8}{27}$
 $\therefore r = \frac{8}{27}$

3. $\sin \theta = \frac{4}{5}$
 $\therefore \cos \theta = \frac{3}{5}$
 $\therefore \tan \theta = \frac{4}{3}$



4. $y = 2 \sin 2x$

Section II
 next page

Section II. Subpart 2016 Mathematics Trial

Question 11.

(a) $16a - 3 + 4a = 20a - 3$

(b) $(2a)^5 - (a)^5 = (2a-1)(4a^4 + 2a^3 + a^2 + a + 1)$

(c) $\frac{8}{2-5} \times \frac{2+5}{2+5} = \frac{6+3}{5} = \frac{9}{5}$

(d) $\lim_{x \rightarrow 8} \frac{(x+8)(x-8)}{(x-8)} = \frac{16}{1} = 16$

(e) $y' = \frac{(e^x + 1)^2 (x) - 2x(e^x + 1)^2}{(e^x + 1)^4}$
 $= \frac{2(e^x + 1)^2 - 2xe^x(e^x + 1)^2}{(e^x + 1)^4}$
 $= \frac{2(1 - xe^x)}{(e^x + 1)^2}$

(f) $a^2 + b^2 = (4+9)^2 = 169$
 $a+b = 13$
 $a-b = 5$

(g) $y' = 2x \cos x - x^2 \sin x$

(h) $\cos x \sin x = \frac{1}{2} \sin 2x$
 $\therefore \cos x \sin x = 0.25$
 $\therefore \sin 2x = 0.5$ (the nearest degree)

Question 12.

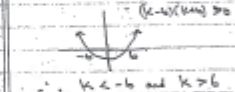
(a) Let $\theta = 0$; $\theta = 0$

(b) $\pi = 10 \times \theta$
 $\therefore \theta = \frac{\pi}{10} = \frac{180}{10} = 18^\circ$

(c) $A = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} (10)^2 \left(\frac{\pi}{10}\right)$
 $= 50 \times \frac{\pi}{10} = 5\pi$

(d) $A = 40\pi - \frac{1}{2} (40)(40) \left(\frac{\pi}{4}\right)$
 $= 40\pi - 20\pi = 20\pi$

(e) $A > 0$ $A = 2\pi$
 $\therefore k^2 - 4 + 12 > 0$
 $k^2 - 8 > 0$
 $(k-2)(k+2) > 0$



$\therefore k < -2$ and $k > 2$
 (f) $2 \sin^2 \theta + \sin \theta = 0$
 $\therefore \sin \theta (2 \sin \theta + 1) = 0$
 $\therefore \sin \theta = 0$ $2 \sin \theta + 1 = 0$
 $\sin \theta = -\frac{1}{2}$

$\theta = 0^\circ, 180^\circ, 225^\circ, 315^\circ$

(g) $AB = \sqrt{(4-0)^2 + (4-1)^2}$
 $= \sqrt{16+9}$
 $= 5$

Section II: cont'd 4012.24 Mathematics Trial Solution

Q.12 (a) cont'd

$$(i) \text{ slope} = \frac{7-4}{0-4} = -\frac{3}{4}$$

$$(ii) 4-y = m(x-0) \quad \text{Using (0,7)}$$

$$4-y = -\frac{3}{4}(x-0)$$

$$4y - 3x = -3x$$

$$\therefore 3x - 4y = 28 = 0$$

$$(iv) d = \left| \frac{0x - 4y - 28}{\sqrt{0^2 + 4^2}} \right|$$

$$d = \left| \frac{3x + 4y + 28}{\sqrt{3^2 + 4^2}} \right|$$

$$d = \left| \frac{27}{5} \right| = \frac{27}{5} \text{ units}$$

$$(v) C(-5,1)$$

$$(vi) A = b \cdot h$$

$$= 5 \times \frac{27}{5} = 27 \text{ units}^2$$

Question 13:

$$(a) (i) y' = 3 \times 2x e^{2x}$$

$$= 6x e^{2x}$$

Question (ii)

$$\int x e^{2x} dx = \frac{1}{2} \int 6x e^{2x} dx$$

$$= \frac{1}{2} \cdot 3 e^{2x} \cdot \frac{1}{2}$$

$$= \frac{3}{4} e^{2x} + C$$

$$(b) f'(x) = 3x^2 + 6x + C$$

$$f'(1) = 0$$

$$\therefore f'(x) = 3x^2 + 6x + C = 0$$

$$C = -3$$

$$\therefore f'(x) = 3x^2 + 6x + 3$$

$$f(x) = x^3 - 3x^2 + 3x + C$$

$$f(2) = 7$$

$$\therefore (2)^3 - 3(2)^2 + 3(2) + C = 7$$

$$8 - 12 + 6 + C = 7$$

$$C = 5$$

$$\therefore f(x) = x^3 - 3x^2 + 3x + 5$$

$$(c) \left[\frac{108}{56} \right] = 23$$

$$(i) P(6) = \frac{5}{22}$$

$$(ii) P(AB) = \frac{10}{22} \times \frac{9}{22}$$

$$= \frac{45}{253}$$

$$(iii) P(88 \text{ or } 88 \text{ or } 99)$$

$$= \frac{45}{253} + \left(\frac{8}{22} \times \frac{7}{22} \right) + \left(\frac{8}{22} \times \frac{4}{22} \right)$$

$$= \frac{16}{253}$$

$$= \frac{16}{253} \quad \left(\frac{16}{253} \right)$$

Section II: cont'd 4012.24 Mathematics Trial Solution

Q.13 (a) cont'd

(d) Note:

$$\$300,000$$

$$4\% \text{ p.a.} \Rightarrow 0.04 \text{ per month}$$

$$1 \text{ year} \Rightarrow 12 \text{ months}$$

$$(i) A_1 = 300,000(1.04)^{12} - M$$

$$(ii) A_1 = A \times (1.04)^n - M$$

$$= (300,000 \times 1.04^{12} - M) \times 1.04^{12}$$

$$= 300,000 \times 1.04^{24} - M(1.04^{12})^2$$

$$\therefore A_2 = 300,000(1.04)^{24} - M(1.04^{12})^2$$

$$A_2 = 300,000(1.04)^{24} - M(1.04^{12})^2$$

$$A_{100} = 300,000(1.04)^{100} - M(1.04^{12})^{100}$$

$$= 300,000(1.04)^{100} - M(1.04^{12})^{100}$$

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$$= 300,000(1.04)^{100} - M(1.04^{12})^{100}$$

Question 14:

$$(a) \int_0^{\pi/2} \sin 2x \, dx$$

$$= \left[-\frac{1}{2} \cos 2x \right]_0^{\pi/2}$$

$$= -\frac{1}{2} \left[\cos \pi - \cos 0 \right]$$

$$= -\frac{1}{2} \left[-1 - 1 \right]$$

$$= -\frac{1}{2} \left[-2 \right]$$

$$= 1$$

$$(b) \text{ New } x^2 - 7x + 10 = 0$$

$$(x-5)(x-2) = 0$$

$$\therefore x = 2, 5$$

$$A = \left| \int_2^5 (x^2 - 7x + 10) \, dx \right|$$

$$= \left| \left[\frac{x^3}{3} - \frac{7x^2}{2} + 10x \right]_2^5 \right|$$

$$= \left| \left(\frac{125}{3} - \frac{175}{2} + 50 \right) - \left(\frac{8}{3} - 14 + 20 \right) \right|$$

$$= \left| \left(\frac{125}{3} - \frac{175}{2} + 50 \right) - \left(\frac{8}{3} - 14 + 20 \right) \right|$$

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#5

Section II, cont'd V12 Math Trial Solution

Q14: cont'd

(c) (i) $y = 3 = \frac{1}{2}(x+2)$

$2y - 1 = x + 2$

When $x = 0$, $2y - 1 = 2$

$y = \frac{3}{2}$

$\therefore T(0, \frac{3}{2})$

(ii) If $m = \tan \theta$

$\therefore \tan \theta = \frac{3}{2}$

$\therefore \theta = \tan^{-1}(\frac{3}{2})$

$\theta = 26.54^\circ$

Angle with y -axis

$90^\circ - 26.54^\circ = 63.46^\circ$

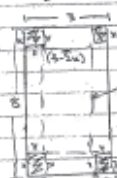
(iv)



$A = \frac{1}{2} \times 3 \times 2$

$= 2.5 \text{ m}^2$

(v)



(1) $V = \text{Length} \times (3-2) \times (2)$

$= (2x - 6x + 6x + 4x) \times 2$

$= 24x - 22x^2 + 4x^3$

Q15

(i) $V = 24x - 22x^2 + 4x^3$

$\frac{dV}{dx} = 24 - 44x + 12x^2$

$\text{Let } \frac{dV}{dx} = 0$

$\therefore 24 - 44x + 12x^2 = 0$

$4(6 - 11x + 3x^2) = 0$

$4(3 - x)(2 - 3x) = 0$

$\therefore x = 3 \text{ or } \frac{2}{3}$

Now $\frac{d^2V}{dx^2} = -44 + 24x$

When $x = 3$, $\frac{d^2V}{dx^2} = -44 + 24(3)$

$= 28$

$\therefore \frac{d^2V}{dx^2} > 0$ Min

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P.S.D.

#6

Section II, cont'd

V12 Math Trial Solution

Question 15:

(a) (i) $y = \log_2(x-1)$

$y = \frac{\ln(x-1)}{\ln 2}$

(ii) $\frac{dy}{dx} = \frac{1}{(x-1)\ln 2}$

$\therefore \frac{dy}{dx} = \frac{1}{(x-1)\ln 2}$

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Q15 cont'd

(c)



(1) In $\triangle ABX$ and $\triangle CYD$

$\angle ABX = \angle CYD$ (alt. angles)

$\angle BAX = \angle DCY$ (alt. angles)

$\therefore \triangle ABX \sim \triangle CYD$ (AA)

$\therefore \frac{AB}{CY} = \frac{BX}{YD}$

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#7

Section II: 4x12 Math Trial Solution 2016

Question 16i

$$y = (x-2)^3(x+1)$$

$$\begin{aligned} \text{(a)} \quad (i) \frac{dy}{dx} &= (x-2)^3 \cdot 1 + (x+1)(x-2)^2 \cdot 1 \\ &= (x-2)^2 [(x-2) + 3(x+1)] \\ &= (x-2)^2 (4x+1) \end{aligned}$$

$$(ii) \frac{d^2y}{dx^2} = 6(x-2)(2x-1)$$

$$\therefore \frac{d^2y}{dx^2} = 0 \quad \text{Solve for } x$$

$$\therefore (x-2)^2(4x+1) = 0$$

$$\therefore x = 2 \text{ or } -1/4$$

Now

$$\text{When } x=2 \quad \frac{d^3y}{dx^3} = 0, \text{ possible P.O.I.}$$

$$\begin{aligned} x = -1/4 \quad \frac{d^2y}{dx^2} &= 6(-9/4)(-3/2) \\ &= \frac{81}{4} \end{aligned}$$

$$\therefore \frac{d^2y}{dx^2} > 0$$

$$\text{Min } (-1/4, -287/25)$$

Now At $x=2$

$$\begin{array}{c|c|c|c} x & 1 & 2 & 3 \\ \hline y'' & -6 & 0 & 30 \end{array}$$

Changes sign

horizontal pt of inflection at

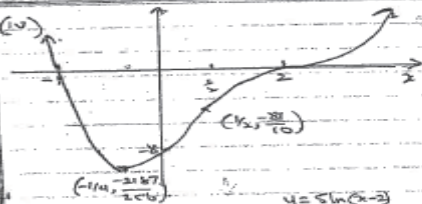
$$(2, 1/5)$$

$$(iii) \text{ Let } \frac{d^2y}{dx^2} = 0 \text{ at } x=2, \pm 1/2$$

$$\begin{array}{c|c|c|c} x & 0 & 1/2 & 1 \\ \hline y'' & 12 & 0 & 6 \end{array}$$

point of inflection $(1/2, -81/16)$

(iv)



$$(b) V = \pi \int_0^5 x^2 dy$$

$$V = \pi \int_0^5 (2 + e^{4x})^2 dy$$

$$= \pi \int_0^5 (4 + 4xe^{4x} + e^{8x}) dy$$

$$= \pi \left[4y + \frac{4x^2}{2} + \frac{e^{8x}}{8} \right]_0^5$$

$$= \pi \left[4y + 20e^{4x} + \frac{1}{8}e^{8x} \right]_0^5$$

$$= \pi \left[(20 + 20e + \frac{5e^2}{2}) - (0 + 20 + \frac{1}{8}) \right]$$

$$= \pi \left[\frac{5e^2}{2} + 20e - \frac{1}{8} \right] \approx 1.3$$

$$(c) (A, B) \quad mx = 6x - x^2$$

$$\therefore x^2 + mx - 6x = 0$$

$$x(x + m - 6) = 0$$

$$\therefore x = 0 \text{ or } 6 - m$$

$$\text{So } A = \int_0^{6-m} (6x - x^2 - mx) dx$$

$$= \left[3x^2 - \frac{x^3}{3} - \frac{mx^2}{2} \right]_0^{6-m}$$

$$= \left[3(6-m)^2 - \frac{(6-m)^3}{3} - \frac{m(6-m)^2}{2} \right] - 0$$

$$= (6-m)^2 \left[3 - \frac{(6-m)}{3} - \frac{m}{2} \right]$$

$$= (6-m)^2 \left[18 - 2(6-m) - 3m \right]$$

$$= \frac{(6-m)^2}{6} (6-m) = \frac{(6-m)^3}{6} \quad \text{END}$$